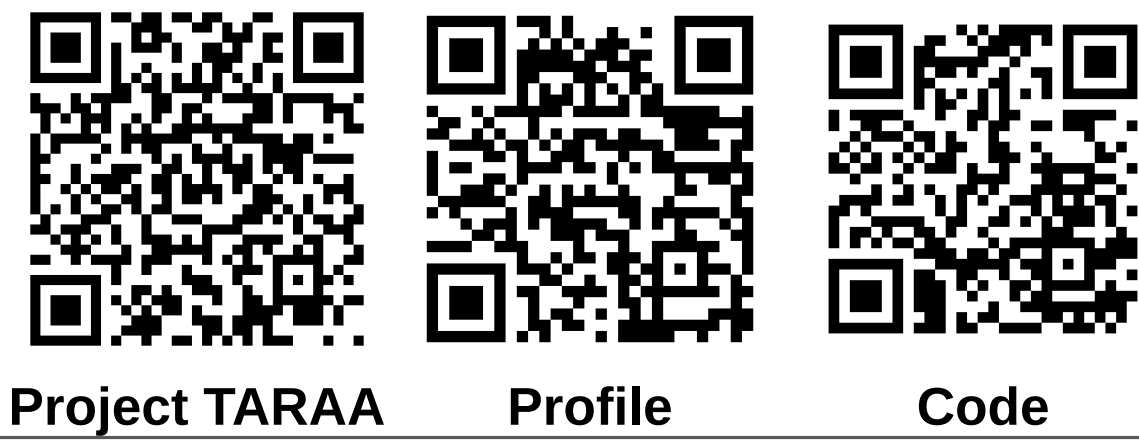


Conserved Immune Topology Improves Pathology Foundation Model Generalization for Cross-Cancer MSI-H Prediction

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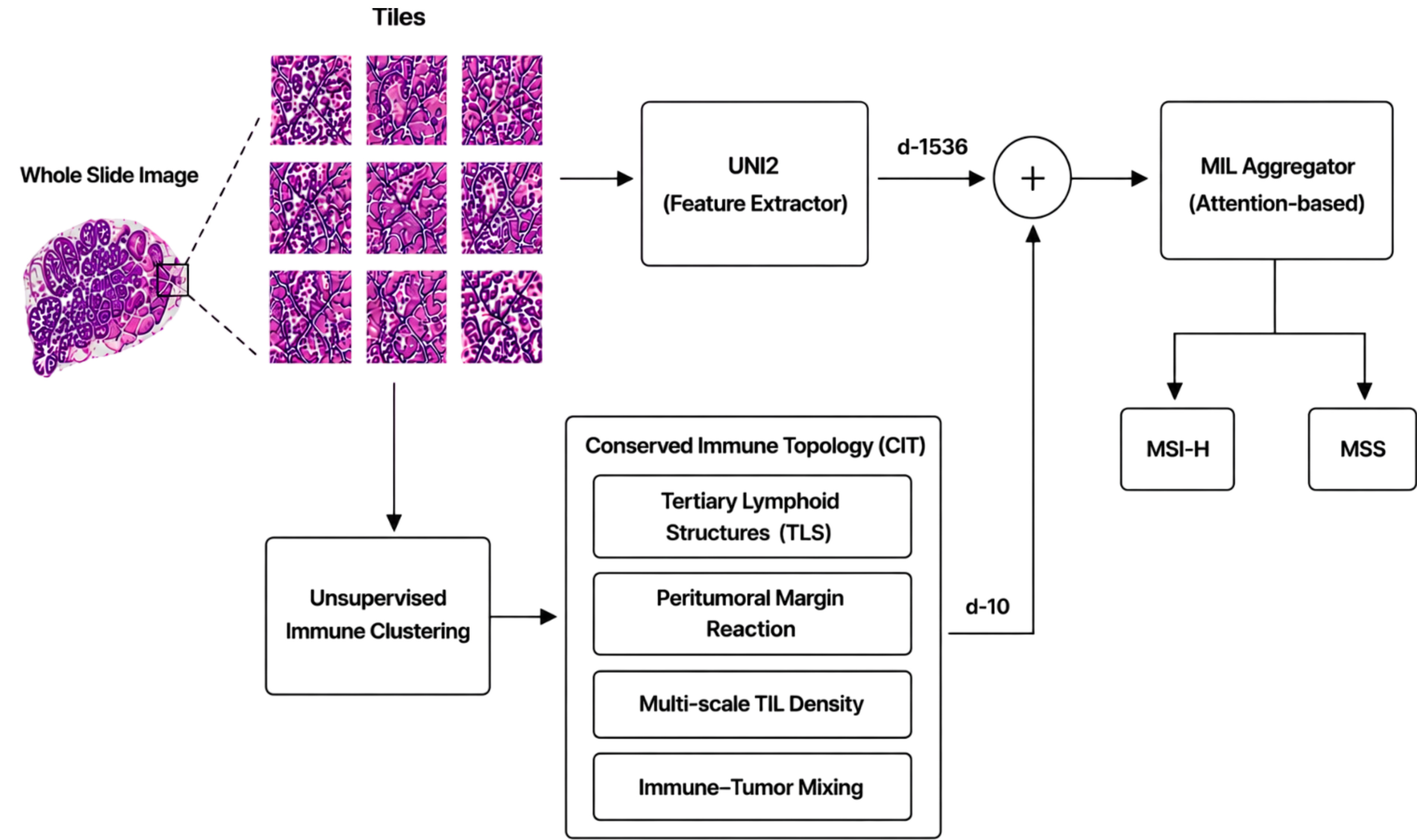
Malmö, Sweden

Problem

Microsatellite instability-high (MSI-H) is an important biomarker for prognosis, treatment selection, and immunotherapy decisions. Pathology foundation model + MIL pipelines perform well within individual cancer cohorts, but cross-cancer transfer remains difficult because organ-specific histology and architecture introduce domain shift.

- Cross-cancer deployment requires representations that are less dependent on organ-specific morphology.
- MSI-H tumors exhibit recurrent immune patterns, including tertiary lymphoid structures, peritumoral lymphocytic reactions, increased TIL density, and immune–tumor mixing across cancer types.
- These conserved spatial immune patterns may provide transferable cues for MSI-H prediction.

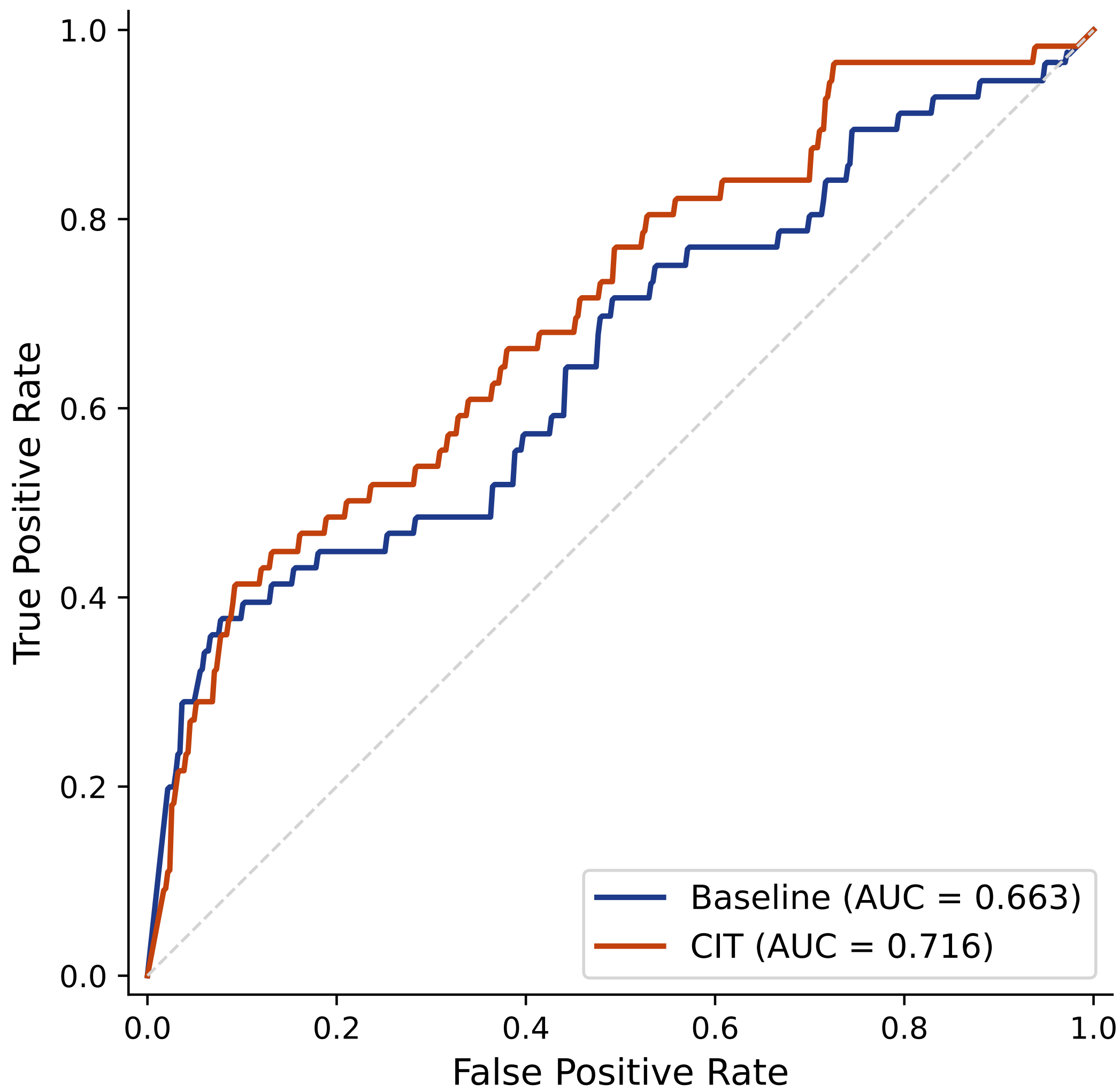
Conserved Immune Topology (CIT) Architecture



- Immune-associated tiles are identified by unsupervised clustering of frozen pathology foundation-model embeddings.
- CIT encodes tertiary lymphoid structures, peritumoral margin reaction, multi-scale immune density, and immune–tumor mixing.
- The resulting 10-dimensional descriptor is concatenated with each frozen tile embedding before MIL aggregation.

Zero-Shot Cross-Cancer ROC: TCGA-STAD

- **TransMIL + CIT** increased MSI-H AUC from **0.6627** to **0.7161** ($\Delta\text{AUC} +0.0534$, $p = 0.003$). The Bonferroni-adjusted 98.33% DeLong confidence interval for the improvement was +0.0111 to +0.0957.



Research Objective

Determine whether conserved immune spatial organization can improve cross-site and cross-cancer MSI-H generalization of pathology foundation models.

- Construct a lightweight Conserved Immune Topology (CIT) representation from frozen foundation-model embeddings and tile coordinates.
- Integrate CIT with standard MIL aggregators without target-domain adaptation.
- Evaluate internal, cross-site, and zero-shot cross-cancer transfer.

Dataset and Evaluation

Three cohorts with molecularly determined MSI labels were used for training and evaluation.

301	105	308
TCGA-COAD	CPTAC-COAD	TCGA-STAD
17.9% MSI-H	22.9% MSI-H	17.5% MSI-H

Training/internal CV: TCGA-COAD

Cross-site test: CPTAC-COAD

Zero-shot cross-cancer test: TCGA-STAD

Internal Validation: TCGA-COAD

Aggregator	Baseline	CIT
ABMIL	0.9337	0.9348
CLAM-SB	0.9469	0.9447
TransMIL	0.9398	0.9567

- CIT preserved within-cohort discrimination, with the highest internal AUC observed for TransMIL + CIT.

Cross-Site Generalization: CPTAC-COAD

Aggregator	Baseline	CIT
ABMIL	0.8089	0.8086
CLAM-SB	0.7989	0.8344
TransMIL	0.7870	0.8277

- CIT improved CLAM-SB and TransMIL under institution and scanner shift, while ABMIL remained unchanged.

Zero-Shot Cross-Cancer Transfer: TCGA-STAD

Model	Base.	CIT	ΔAUC	p
ABMIL	0.5681	0.6313	+0.0632	< 0.001
CLAM-SB	0.6242	0.6414	+0.0172	0.037
TransMIL	0.6627	0.7161	+0.0534	0.003

- CIT increased AUC across all three MIL aggregators; ABMIL and TransMIL remained statistically significant after Bonferroni correction.

Takeaway

- Conserved immune spatial organization provides transferable information for MSI-H prediction across cancer types.
- CIT improves generalization without target-domain adaptation, cell annotations, or changes to the frozen foundation model.
- The strongest practical effect appears under distribution shift, particularly zero-shot cross-cancer transfer.